

# Week 17: Mongoose ODM

## ***1. Schema and Model Definition***

Mongoose is an Object Data Modeling (ODM) library used with MongoDB in Node.js applications. It helps define the structure of data using schemas. A schema specifies fields, data types, default values, and constraints for documents in a collection. Models are then created from schemas and used to interact with the database. This makes database operations more organized, structured, and easier to manage.

## ***2. Data Validation***

Data validation ensures that only correct and properly formatted data is stored in the database. Mongoose provides built-in validation features such as required fields, minimum length, maximum length, unique values, and custom validators. Validation reduces errors and improves data consistency. It also helps maintain application reliability.

## ***3. CRUD with Mongoose***

Mongoose simplifies CRUD operations such as Create, Read, Update, and Delete. Methods like `save()`, `find()`, `findById()`, `findByIdAndUpdate()`, and `findByIdAndDelete()` are commonly used. These methods make database interaction easier and reduce development complexity. CRUD operations are essential for building dynamic applications.

## ***4. Relationships (Populate, Ref)***

Relationships between collections can be managed using references (`ref`) in Mongoose. The `populate()` method is used to fetch related data from another collection. This helps connect related entities such as users, posts, comments, or messages. It improves efficiency in handling relational-like data.

## ***5. Middleware in Mongoose***

Mongoose supports middleware functions such as pre and post hooks. These hooks are used to perform actions before or after operations like save, update, or delete. Middleware is useful for hashing passwords, logging, and validation tasks.

## ***Conclusion***

This week focused on Mongoose ODM. By learning schemas, models, validation, CRUD operations, relationships, and middleware, stronger database integration and backend development skills were developed for MERN stack applications.